

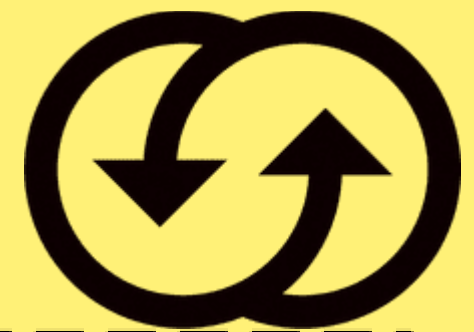
Book Search App



HITESH
23H51A66G5

KRISHNA
23H51A66H2

MedhaSri
23H51A66J3



Contents



01

Introduction

02

Objectives

03

Tech Stack

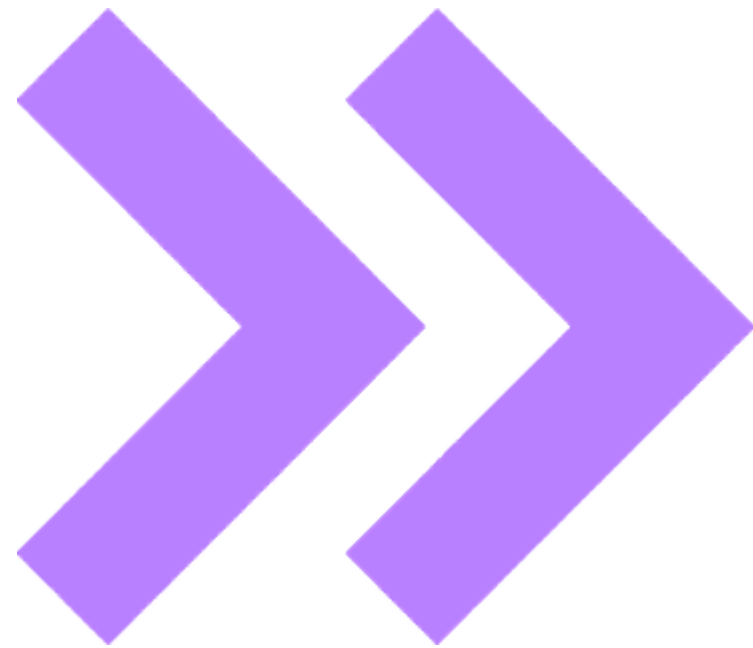
04

App Architecture

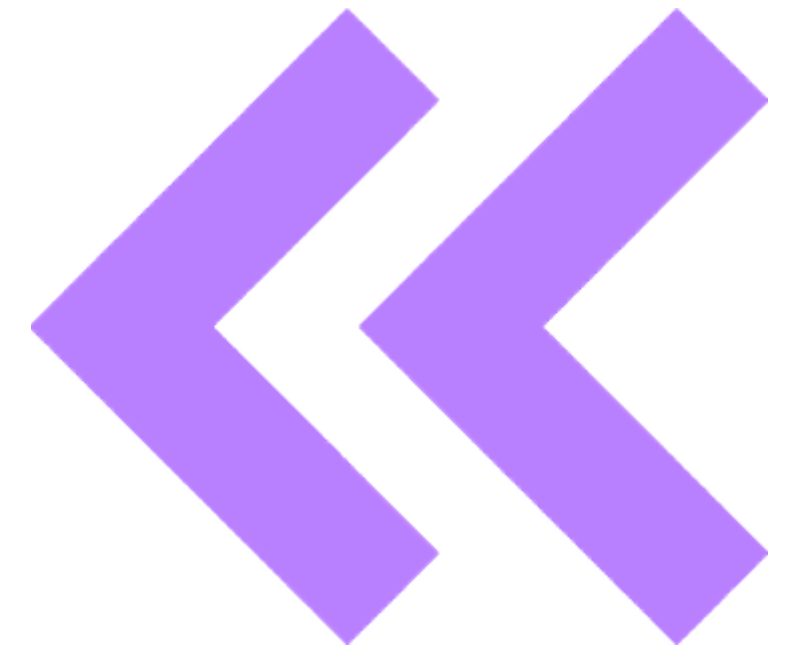
05

UI Design Philosophy

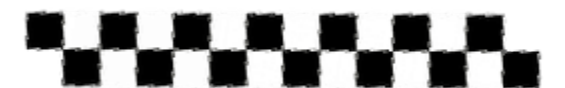




01



Introduction





Book Search App Overview



App Introduction

The Book Search App is a lightweight Flutter prototype designed for quick book discovery. It provides a clean, light-themed UI with minimal aesthetics, ensuring a distraction-free user experience.

01

Prototype Purpose

This app serves as a usability-first prototype, focusing on core search functionality. It aims to validate intuitive search flows and minimal UI principles within Flutter, setting the stage for future enhancements.

02

Project Purpose and Vision

Core Purpose

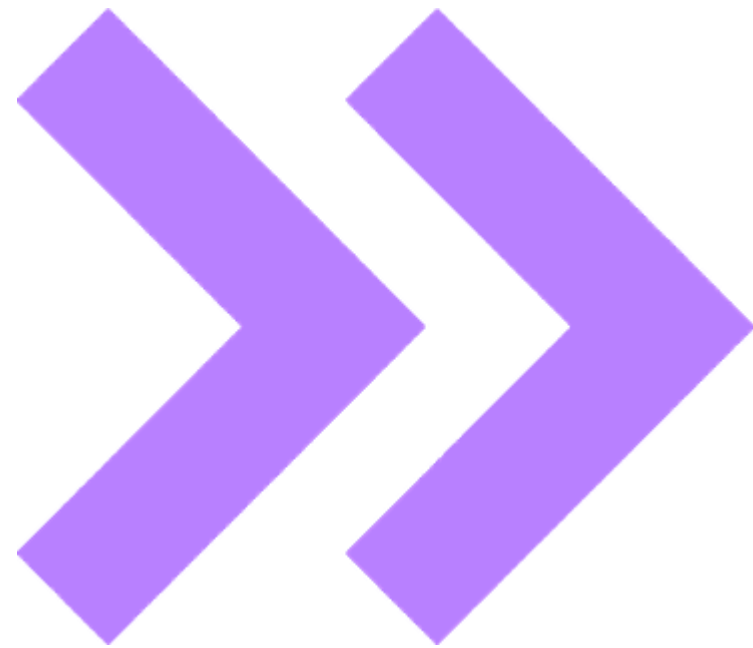
The primary purpose of the Book Search App is to offer a simple, intuitive book finder tool. It prioritizes clarity and speed, ensuring users can quickly locate books without unnecessary complexity.

Design Philosophy

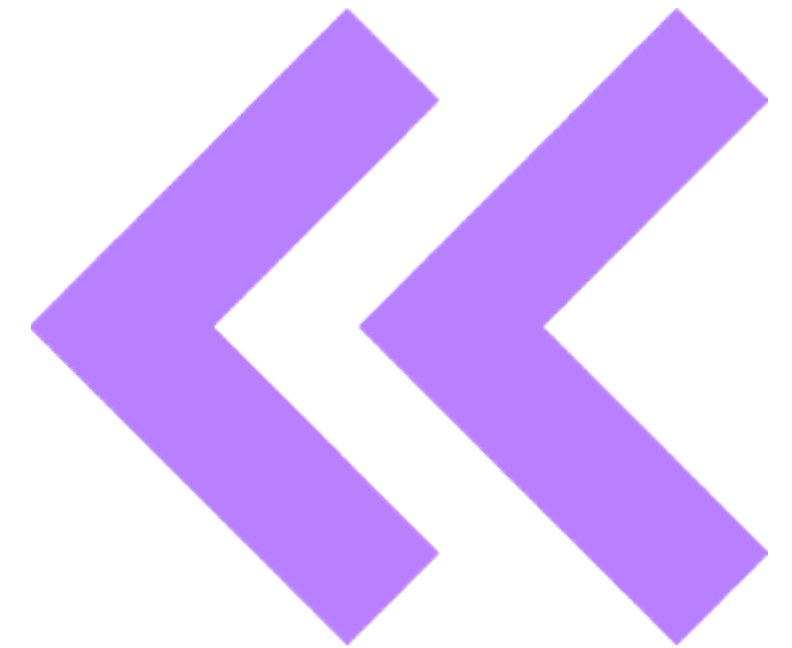
The app adheres to a minimal design philosophy, using soft contrasts and whitespace to create a calm environment. This approach ensures users remain focused on the search functionality.

Future Vision

While currently a prototype, the app is designed with future scalability in mind. It lays the groundwork for integrating real-time data and additional features like book details and bookmarking.



02



Objectives

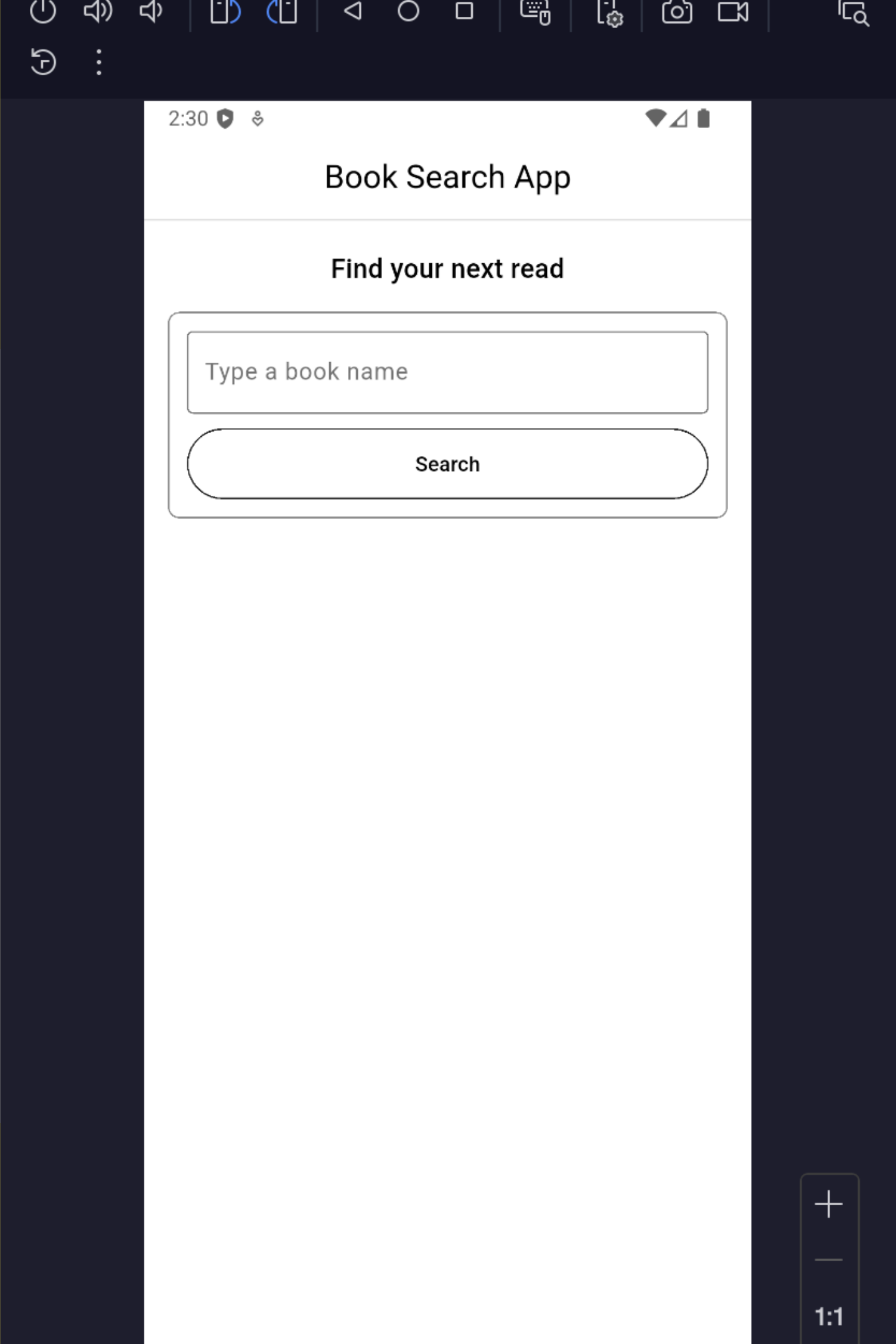


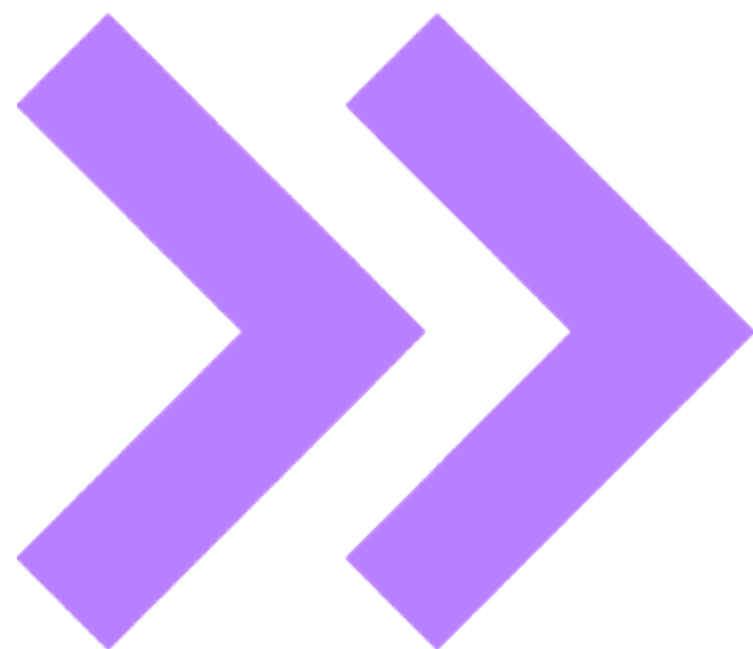


Core Development Goals

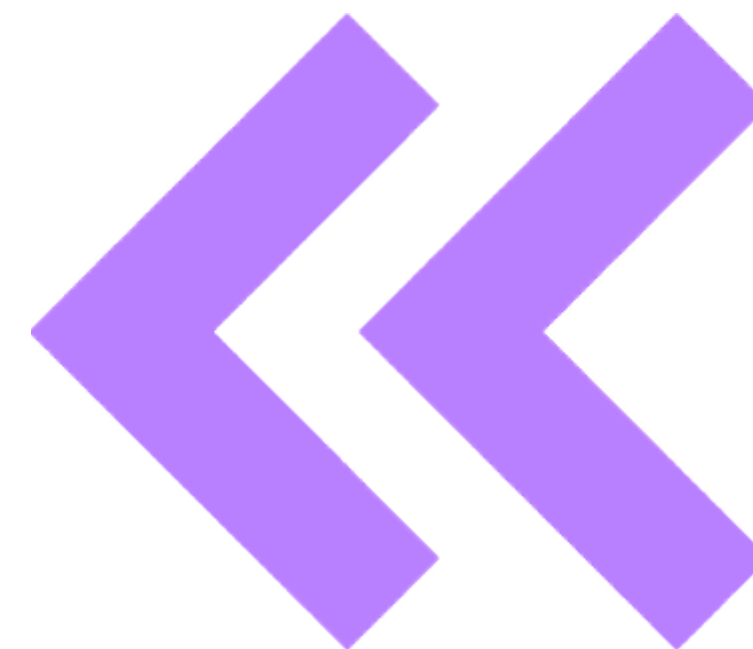
Development Objectives

The main objectives of this project are to build a two-page Flutter app, test search functionality with dummy data, and refine a minimal UI for a smooth user experience.





03



Tech Stack





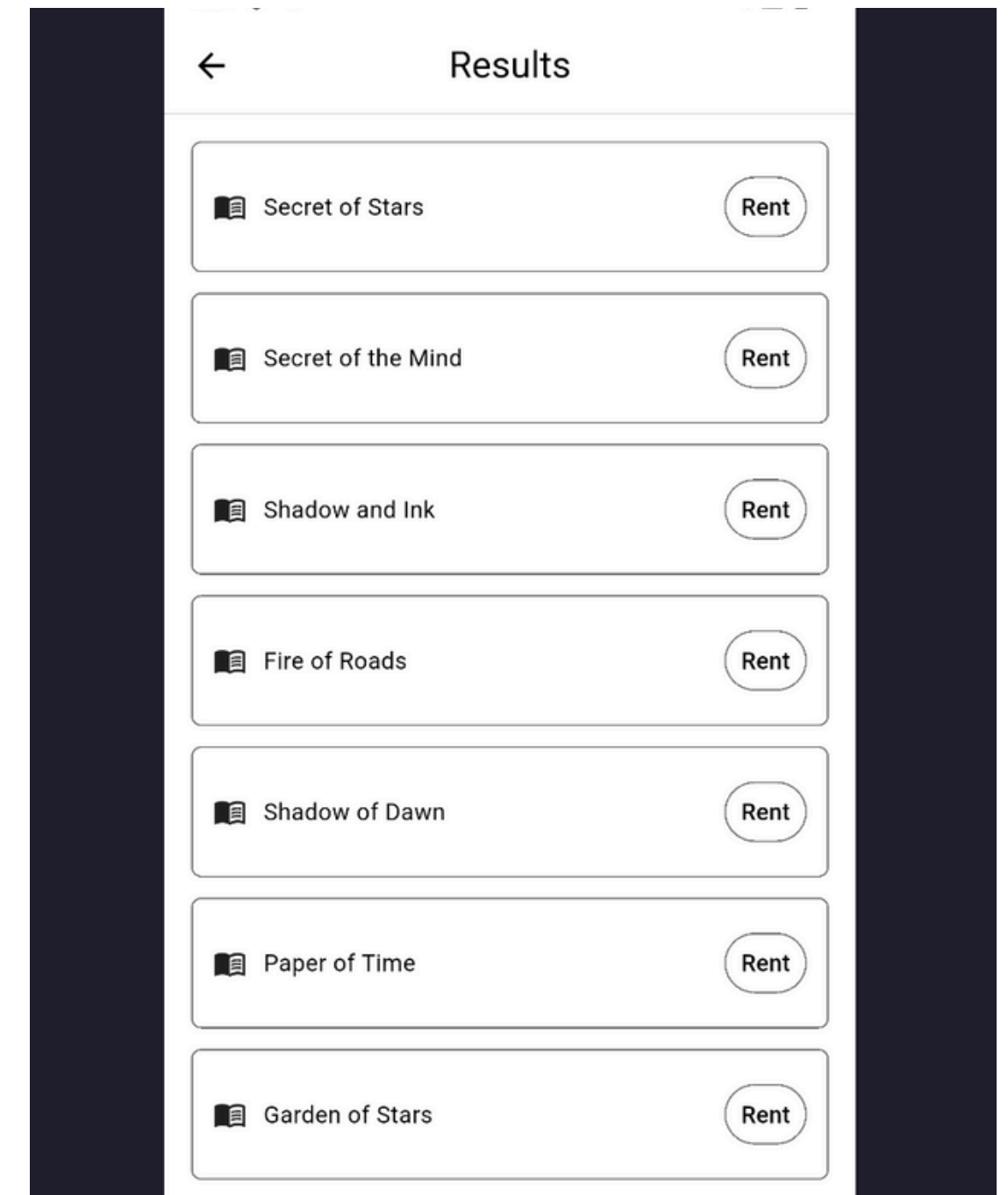
Flutter and Dart Foundation-----

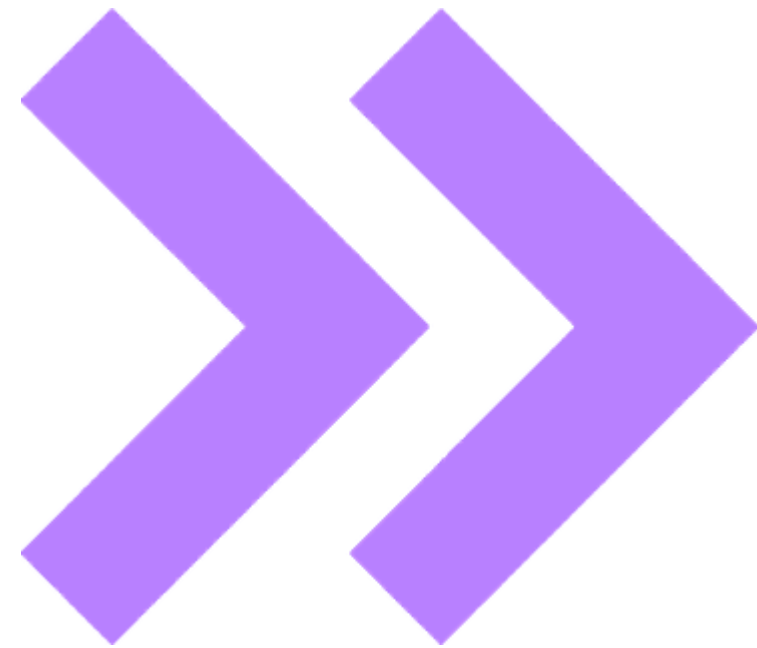
Flutter SDK

Flutter is used as the primary framework for building the app. It enables cross-platform UI development, ensuring the app runs smoothly on both iOS and Android devices.

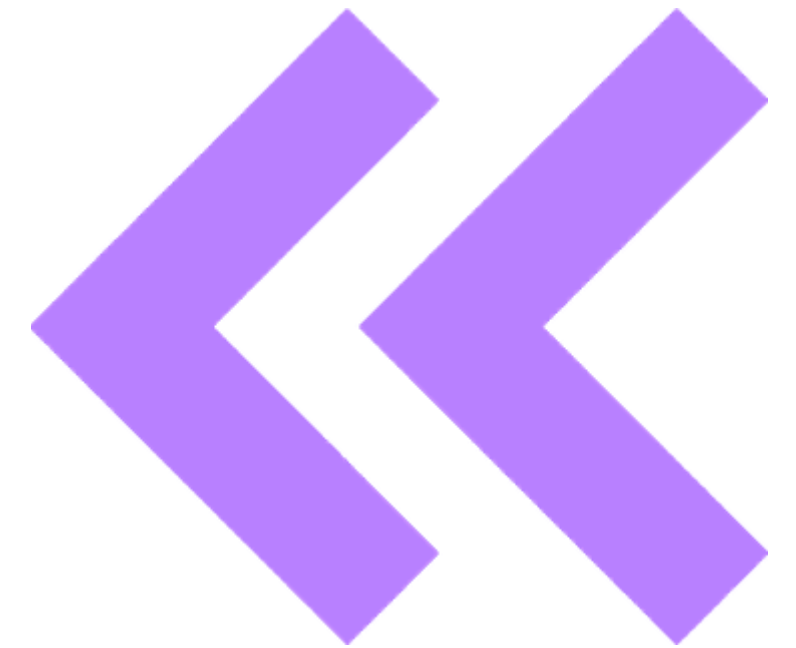
Dart Programming

Dart is utilized for its reactive and type-safe capabilities. It allows for efficient widget composition and smooth animations, enhancing the overall user experience.





04



App Architecture





Two-Page Navigation Flow

Home Page

The Home Page features a centered search bar and an animated search button. Tapping the button triggers navigation to the Results Page.

Results Page

The Results Page displays a list of randomly generated book titles in bordered containers. It includes a Back button for easy navigation.

Navigation Mechanics

The app uses Flutter's routing system to manage transitions between the Home Page and Results Page, ensuring smooth and responsive navigation.



Animation for Modern UX

Subtle Animations

01

The search button features a subtle scaling animation on tap, providing tactile feedback and enhancing the user experience without overwhelming the interface.

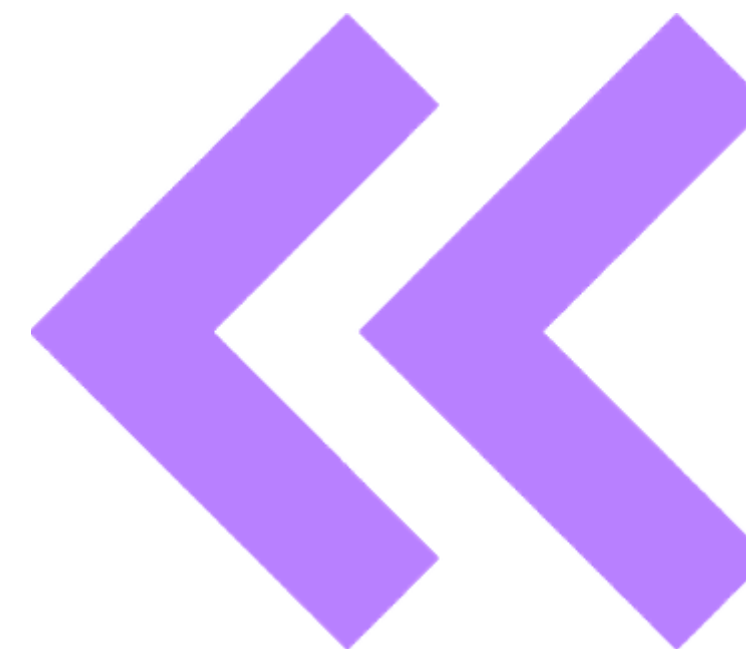
Responsive Design

02

Animations are carefully designed to be smooth and responsive, ensuring that user interactions feel immediate and satisfying, reinforcing the modern UX philosophy.



06



Code



main.dart

```
import 'package:flutter/material.dart';
import 'package:book_search_app/pages/home_page.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    final base = ThemeData.light();
    // Minimal light theme: white background, dark text, simple borders.
    final theme = base.copyWith(
      scaffoldBackgroundColor: Colors.white,
      colorScheme: const ColorScheme.light(
        primary: Colors.black,
        secondary: Colors.black,
        surface: Colors.white,
        onPrimary: Colors.white,
        onSecondary: Colors.white,
        onSurface: Colors.black,
      ),
      appBarTheme: const AppBarTheme(
        backgroundColor: Colors.white,
        foregroundColor: Colors.black,
        elevation: 0,
        centerTitle: true,
      ),
      textTheme: base.textTheme.apply(
        bodyColor: Colors.black,
        displayColor: Colors.black,
      ),
      inputDecorationTheme: const InputDecorationTheme(
        border: OutlineInputBorder(
          borderSide: BorderSide(color: Colors.black54, width: 1),
        ),
        enabledBorder: OutlineInputBorder(
          borderSide: BorderSide(color: Colors.black54, width: 1),
        ),
        focusedBorder: OutlineInputBorder(
          borderSide: BorderSide(color: Colors.black87, width: 1.2),
        ),
        hintStyle: TextStyle(color: Colors.black54),
      ),
      dividerColor: Colors.black26,
    );

    return MaterialApp(
      title: 'Book Search App',
      debugShowCheckedModeBanner: false,
      theme: theme,
      home: const HomePage(),
    );
  }
}
```

home_page.dart

```
import 'package:flutter/material.dart';
import 'results_page.dart';

class HomePage extends StatefulWidget {
  const HomePage({super.key});

  @override
  State<HomePage> createState() => _HomePageState();
}

class _HomePageState extends State<HomePage> with TickerProviderStateMixin {
  final TextEditingController _controller = TextEditingController();
  late final AnimationController _pressController;
  late final Animation<double> _scale;

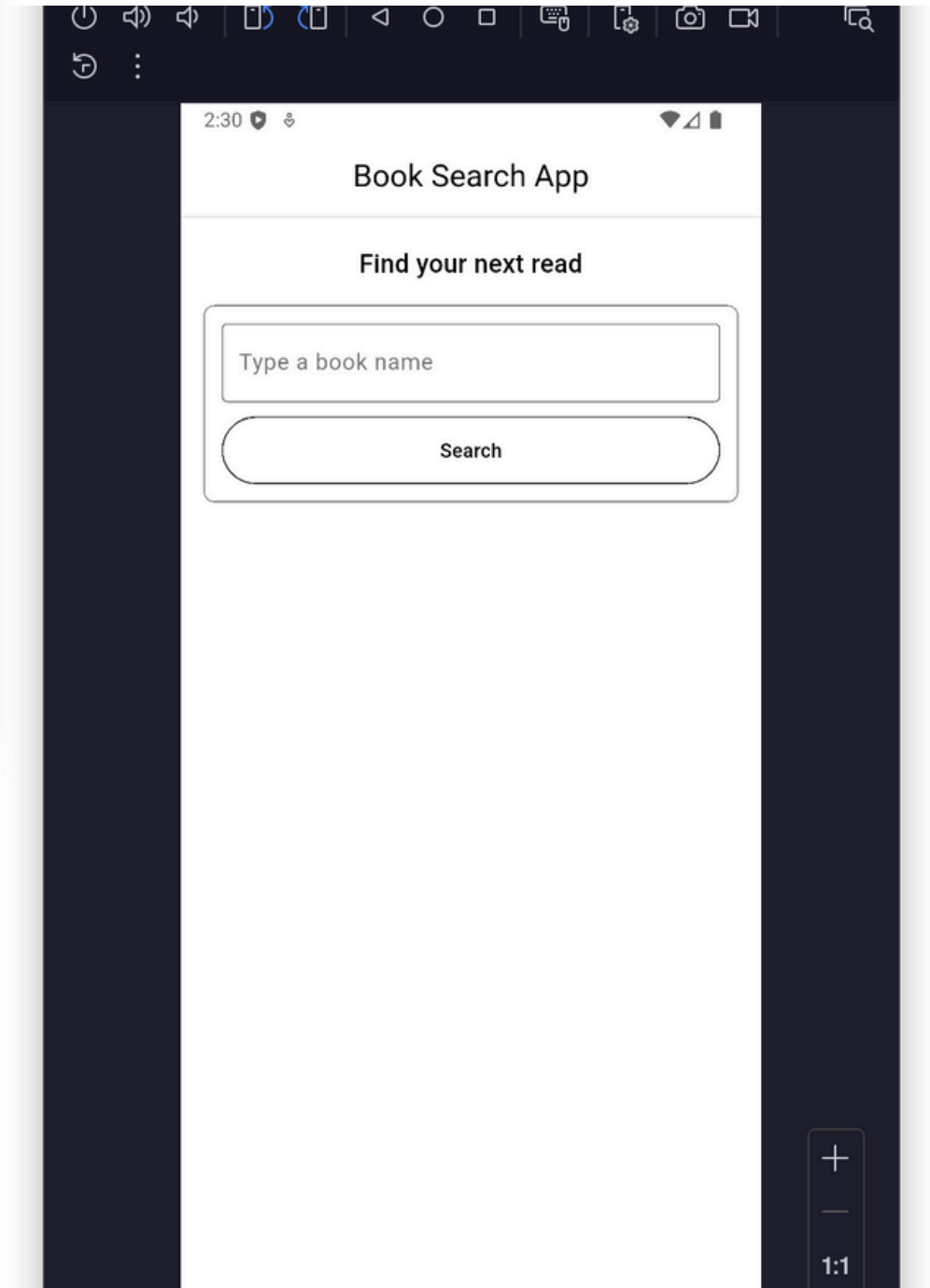
  late final AnimationController _introController;
  late final Animation<double> _fadeTitle;
  late final Animation<Offset> _slideTitle;
  late final Animation<double> _fadeContent;
  late final Animation<Offset> _slideContent;

  Set<String> _rented = <String>{};
}
```

home_page.dart

```
@override
void initState() {
  super.initState();
  // Button press animation
  _pressController = AnimationController(
    vsync: this,
    duration: const Duration(milliseconds: 160),
    reverseDuration: const Duration(milliseconds: 140),
  );
  _scale = Tween<double>(begin: 1.0, end: 0.94).animate(
    CurvedAnimation(parent: _pressController, curve: Curves.easeOut, reverseCurve: Curves.easeIn),
  );

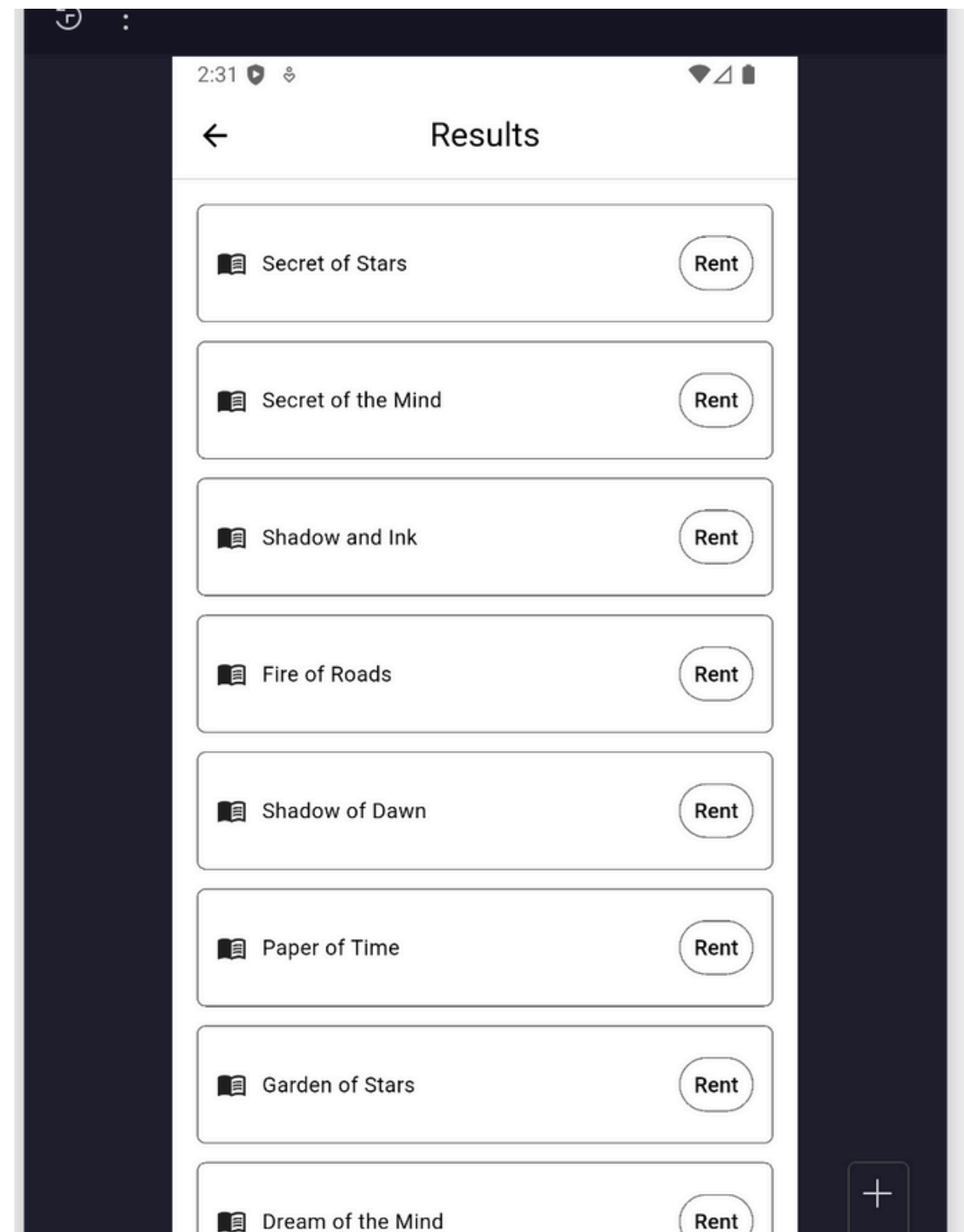
  // Intro animations
  _introController = AnimationController(
    vsync: this,
    duration: const Duration(milliseconds: 600),
  );
  _fadeTitle = CurvedAnimation(parent: _introController, curve: const Interval(0.0, 0.5, curve: Curves.easeOut));
  _slideTitle = Tween<Offset>(begin: const Offset(0, -0.08), end: Offset.zero).animate(
    CurvedAnimation(parent: _introController, curve: const Interval(0.0, 0.6, curve: Curves.easeOutCubic)),
  );
  _fadeContent = CurvedAnimation(parent: _introController, curve: const Interval(0.2, 1.0, curve: Curves.easeOut));
  _slideContent = Tween<Offset>(begin: const Offset(0, 0.08), end: Offset.zero).animate(
    CurvedAnimation(parent: _introController, curve: const Interval(0.2, 1.0, curve: Curves.easeOutCubic)),
  );
}
```

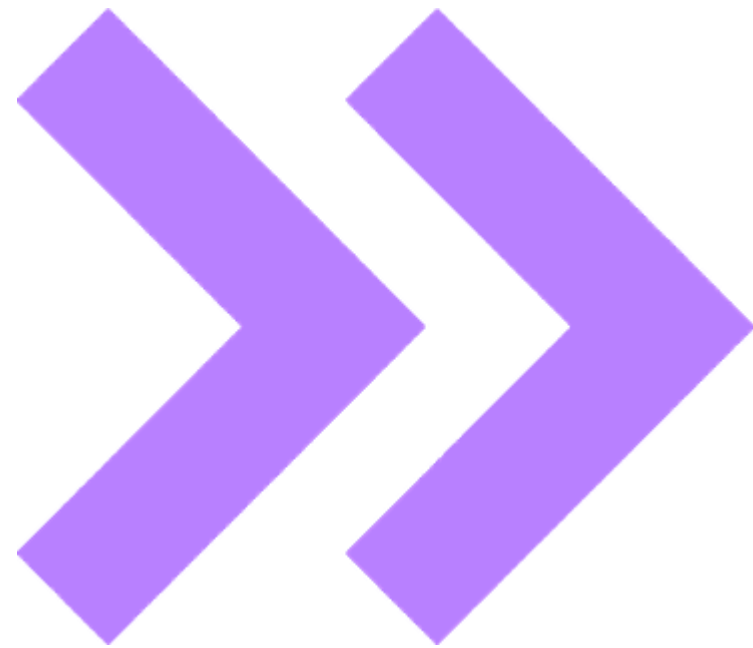


results_page.dart

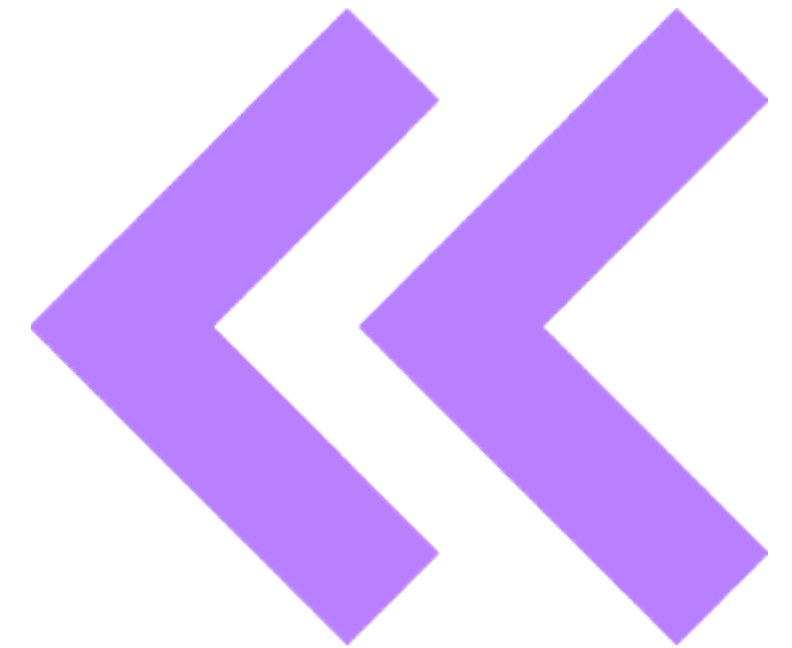
```
List<String> _generateFakeBooks(String q) {  
  final titles1 = [  
    'Shadow', 'Memory', 'Echo', 'Secret', 'Light', 'Night', 'Dream', 'Garden', 'River', 'City',  
    'Story', 'Path', 'Stone', 'Wind', 'Sea', 'Sky', 'Fire', 'Glass', 'Paper', 'Code'  
  ];  
  final titles2 = [  
    'of Time', 'in the Rain', 'of Tomorrow', 'and Dust', 'of Silence', 'of Dreams',  
    'of the Mind', 'of Dawn', 'and Shadows', 'of the Lost', 'of Hope', 'of Winter',  
    'of Summer', 'of Stars', 'of Maps', 'and Mirrors', 'and Ink', 'of Secrets',  
    'of Roads', 'and Stories'  
  ];  
}
```

snappify.com





07



Conclusion





Prototype Achievements

Successful Demonstration

The Book Search App successfully demonstrates Flutter navigation, stateless UI composition, and purposeful animation, validating the minimalist design approach.

Iterative Potential

The app provides a solid foundation for iterative enhancement, allowing for the gradual addition of advanced features while maintaining usability.

Functional Value

The prototype delivers functional value through a simple two-page structure, ensuring users can quickly search for books and view results.

User-Centric Design

By keeping cognitive load low, the app ensures a user-centric experience, making it an ideal starting point for further development.



THANK YOU

